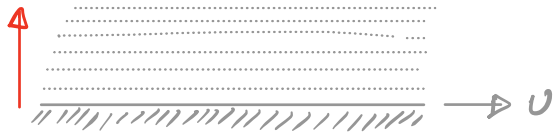


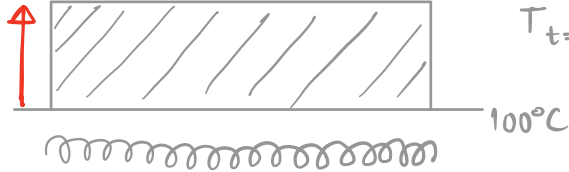
Ex:- direction of momentum transport



This transport of momentum is "molecular" and "diffusive" in nature.

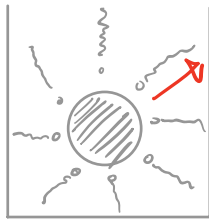
Ex:-

diffusive transport of heat energy (conduction)



$T_{t=0} = 25^\circ\text{C}$ (Initially the slab is at room temp.)

Ex:-



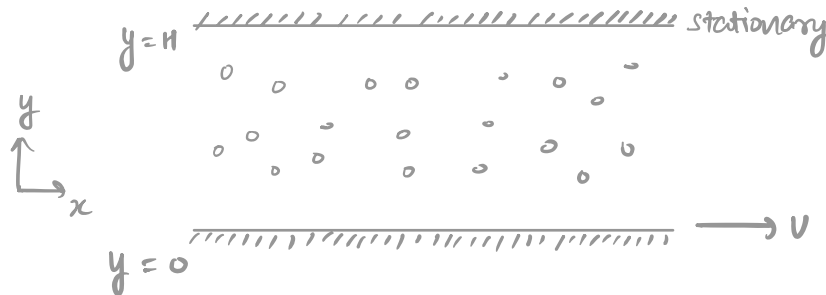
diffusive mass transfer

Therefore Transport Phenomena



(Momentum, Energy, Mass)

(Q.)



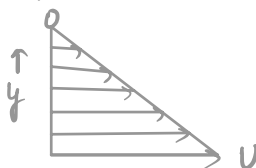
(*) Initial condition ($t=0$)

$$\forall x = 0 \quad \forall (x, y)$$

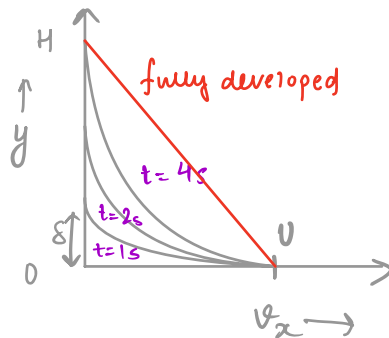
(*) Boundary Conditions

$$\left. \begin{array}{l} y=0, v_x = u \quad \forall t \\ y=H, v_x = 0 \quad \forall t \end{array} \right\} \begin{array}{l} \text{no} \\ \text{slip} \end{array}$$

→ At steady state, velocity profile will be



→ logically the developing phase can be shown as follows:-



$\delta(t)$: Momentum diffusion length scale.

δ : roughly the length scale upto which molecules are in motion or upto which momentum has transferred.

→ Cauchy - momentum balance (STUDY)

→ for newtonian fluids $\dot{\gamma}$ (shear rate)

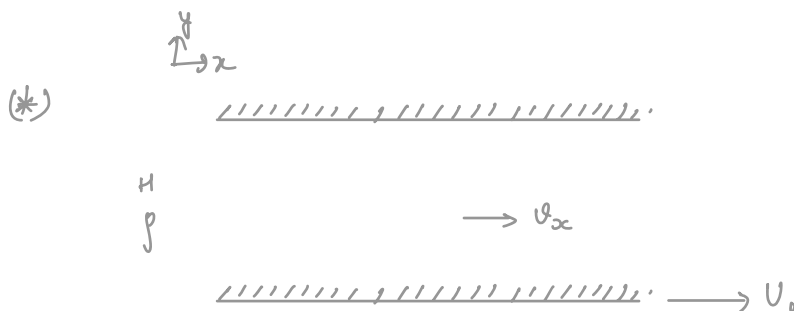
$$\tau_{xy} = -\eta \left(\frac{\partial v_x}{\partial y} \right)$$

Now writing governing equation for the above problem (2D, incompressible)

$$\rho \left(\underbrace{\frac{\partial v_x}{\partial t}}_0 + \underbrace{v_x \frac{\partial v_x}{\partial x}}_0 + \underbrace{v_y \frac{\partial v_x}{\partial y}}_0 + \underbrace{v_z \frac{\partial v_x}{\partial z}}_0 \right) = \underbrace{-\frac{\partial p}{\partial x}}_0 + \underbrace{\mu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right)}_0 + \underbrace{\rho g_x}_0$$

(fully developed along x-direction) (not pressure driven) (body force only) in z-direction

$$\boxed{\rho \frac{\partial v_x}{\partial t} = \mu \frac{\partial^2 v_x}{\partial y^2}}$$



Our target is to find

$$v_x(y, t)$$

↓
velocity profile

Assumptions

Laminar flow
2D flow

fully developed flow
No slip

$$\text{For newtonian fluid} \Rightarrow \tau = -\mu \frac{\partial v_x}{\partial y}$$

We know

$$\rho \frac{\partial v_x}{\partial t} = \mu \frac{\partial^2 v_x}{\partial y^2}$$

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

ν : kinematic viscosity /
momentum diffusivity

$$= \frac{\mu}{\rho}$$

Boundary conditions

$$y=0, v_x = U_0, \forall t$$

$$y=H, v_x = 0, \forall t$$

Initial condition

$$t=0, v_x = 0, \forall y$$

Characteristic Value (order of magnitude analysis)

→ order of magnitude of v_x : U_0

→ order of magnitude of time: t^*

→ order of magnitude of ∂v_x : U_0

→ order of magnitude of ∂y : H

$$\frac{U_0}{t^*} \sim \nu \left(\frac{U_0}{H^2} \right)$$

$$t^* \sim \frac{H^2}{\nu}$$

characteristic time

We have a time scale associated with
momentum transport $\sim H^2/\nu$

If time of observation $\gg t^*$, then
we have steady state.

After analysis: $v_x \sim U_0$, $y \sim H$, $t \sim \frac{H^2}{\nu}$

Now we will perform non-dimensionalization

$$v_x^* = \frac{v_x}{U_0}$$

$$y^* = \frac{y}{H}$$

$$t^* = \frac{t}{(H^2/\nu)}$$

substituting in

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}$$

$$\left(\frac{U_0}{H^2/\nu} \right) \frac{\partial v_x^*}{\partial t^*} = \frac{\nu U_0}{H^2} \frac{\partial^2 v_x^*}{\partial y^{*2}}$$

$$\Rightarrow \frac{\partial v_x^*}{\partial t^*} = \frac{\partial^2 v_x^*}{\partial y^{*2}}$$

(*) As long as we are close to $t^* = \frac{t}{\frac{H^2}{\nu}}$ then order of magnitude of LHS = 1

(*) If t is very large as compared $\frac{H^2}{\nu}$ then we drop the term $\frac{\partial v_x^*}{\partial t^*}$ as $\partial t^* \gg \partial v_x^*$ thus giving us $\frac{\partial^2 v_x^*}{\partial y^{*2}} = 0$ (STEADY STATE)

↓
logical !!

Boundary conditions

$$\left\{ \begin{array}{lll} y^* = 0 & v_x^* = 1 & \forall t^* \\ y^* = 1 & v_x^* = 0 & \forall t^* \end{array} \right\} \text{ Boundary Conditions}$$

$$\{ t^* = 0, v_x^* = 0, \forall y^* \} \text{ Initial condition}$$

The following equation (proposed solution):-

$$v_x(y, t) = \underbrace{v_{ss}(y)}_{\text{steady state}} + \underbrace{v_t(y, t)}_{\text{unsteady state}}$$

→ All variables are non-dimensional (for convenience we drop the * symbol).

$$\left. \begin{array}{l} (*) \frac{\partial^2 v_{ss}}{\partial y^2} = 0 \\ v_{ss}(y) = ay + b \end{array} \right\} \text{ steady state part.}$$

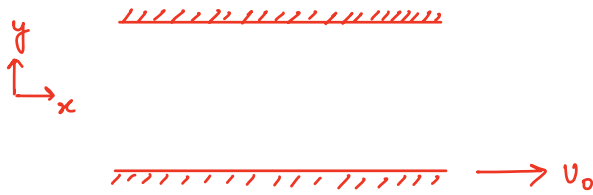
Applying boundary condition

$$y = 0, v_{ss} = 1$$

$$y = 1, v_{ss} = 0$$

$$v_{ss}(y) = 1 - y$$

PRITHVI RAJ SURYAVANSHI (20CH3FP52)



$$\frac{\partial v_x}{\partial t} = \frac{\partial^2 v_x}{\partial y^2}$$

$$v_x = v_{ss} + v_t$$

Our aim is to satisfy

- ① Governing eq'n
- ② Boundary eq'n
- ③ Initial condition

SS

$$\frac{\partial^2 v_{ss}}{\partial y^2} = 0 \Rightarrow v_{ss} = ay + b$$

$$v_{ss} = 1 - y \quad (\text{The B.C. for } v_{ss} \text{ \& Total velocity } (v_x) \text{ will remain the same})$$

$$\therefore v_x = v_{ss} + v_t$$

$$v_x(y, t) = 1 - y + v_t(y, t)$$

Now substituting in governing eq'n

$$\frac{\partial v_x}{\partial t} = 0 - 1 + \frac{\partial v_t}{\partial t}$$

$$\frac{\partial^2 v_x}{\partial y^2} = \frac{\partial^2 v_t}{\partial y^2}$$

We know from NS

$$\frac{\partial v_x}{\partial t} = \frac{\partial^2 v_x}{\partial y^2}$$

$$\frac{\partial v_t}{\partial t} = \frac{\partial^2 v_t}{\partial y^2}$$

⊛ $\frac{\partial v_t}{\partial t} = \frac{\partial^2 v_t}{\partial y^2} \rightarrow v_t = v_{x,t}(y, t)$

B.C

① $y = 0 \Rightarrow v_x = 1$ → non-dimensional term (all variables are non-dimensional)
 $\Rightarrow 1 - y + v_t = 1$
 $\Rightarrow 1 - 0 + v_t = 1 \Rightarrow v_t = 0, \forall t$ $t \sim \frac{H^2}{\nu}$

② $y = 1 \Rightarrow v_x = 0$
 $\Rightarrow 1 - y + v_t = 0$

$$\Rightarrow v_t = 0, \quad \forall t$$

I.C

① $t=0$, $v_x = 0 \quad \forall y$
 $1-y + v_t = 0$
 $v_t = y - 1$ ↗ not physically realizable

Now applying separation of variable: $v_t = f(y)g(t)$

↳ substituting in eq'n (*)

$$f \frac{\partial g}{\partial t} = g \frac{\partial^2 f}{\partial y^2}$$

$$f g' = g f''$$

$$\frac{g'}{g} = \frac{f''}{f} = c$$

only func'n
of t

only func'n
of y

⇒ only possible when both
are constant

$$\frac{1}{g} \frac{dg}{dt} = c$$

$$\Rightarrow \frac{dg}{g} = c dt \Rightarrow \ln g = ct + d$$

$$g = e^{ct+d}$$

$$g = A e^{ct}$$

c = always a negative number
 so that after a long time
 $g \rightarrow 0$ (converges)

Therefore, writing it in a new form

$$\frac{1}{g} \frac{dg}{dt} = -c^2$$

$$g = A e^{-c^2 t}$$

Now, $\frac{g'}{g} = \frac{f''}{f} = -c^2$

$$\frac{1}{f} \frac{d^2 f}{dy^2} = -c^2 \Rightarrow \frac{d^2 f}{dy^2} + c^2 f = 0$$

$$\Rightarrow f = B \sin(cy) + D \cos(cy)$$

$$(*) \quad y=0, \quad \forall t=0 \quad \forall t$$

$$f(y) g(t) = 0 \quad \forall t$$

$$\Rightarrow f(y) = 0, \quad g(t) \text{ can be zero for some } t, \text{ but not all } t.$$

$$y=1, \quad f(y) = 0 \quad \forall t$$

$$\text{At } y=0, y=1, \quad f=0$$

$$\text{At } y=0 \Rightarrow 0 = 0 + D \Rightarrow D=0$$

$$\text{At } y=1 \Rightarrow 0 = B \sin c \Rightarrow \sin c = 0 \Rightarrow c = n\pi$$

$$v_t(y, t) = \sum A_n e^{-c_n^2 t} B_n \sin(c_n y) \quad \leftarrow \text{general solution}$$

$$= \sum D_n e^{-c_n^2 t} \sin(c_n y)$$

$$v_x = v_{ss} + v_t$$

$$= 1-y + \sum_{n=-\infty}^{\infty} D_n e^{-c_n^2 t} \sin(c_n y)$$

##) 02/02/2023

////////////////////

//////////////////// $\rightarrow v_0$

$$v_{sc}(y^*, t^*) = \left(\sum_{n=-\infty}^{\infty} D_n \exp(-n^2 \pi^2 t^*) \sin(n\pi y^*) \right) + (1-y)$$

As of now we have applied B.C's but have not applied I.C.

So now applying Initial Conditions:-

$$t=0, v=0$$

$$\bar{y}^{-1} = \sum_{n=-\infty}^{\infty} D_n \sin(n\pi y^*)$$

coupling negative & positive counterparts

$$\bar{y}^{-1} = \sum_{n=1}^{\infty} \phi_n \sin(n\pi y^*)$$

y^* is written as y after this step

$$(\bar{y}-1) \sin(m\pi y) = \sum_{n=1}^{\infty} \phi_n \sin(n\pi y) \sin(m\pi y)$$

$$\int_0^1 (\bar{y}-1) \sin(m\pi y) dy = \sum_{n=1}^{\infty} \int_0^1 \sin(n\pi y) \sin(m\pi y) dy$$

$0 (m \neq n)$

only surviving term $\Rightarrow n=m$

$$\int_0^1 (\bar{y}-1) \sin(n\pi y) dy = \phi_n \int_0^1 \sin^2(n\pi y) dy$$

$$\phi_n = \frac{-2}{n\pi} \quad (\text{after solving})$$

$$u_x^*(y^*, t^*) = \left[\sum_{n=1}^{\infty} \left(\frac{-2}{n\pi} \right) \exp(-n^2 \pi^2 t^*) \sin(n\pi y^*) \right] + (1-\bar{y})$$

(#) A function of the form $P_n = \sin(c_n y)$ will satisfy
 $P_m = \sin(c_m y)$

$$P_m \frac{d^2 P_n}{dy^2} + c_n^2 P_n P_m = 0$$

Integrating

$$\int_0^1 P_m \frac{d^2 P_n}{dy^2} dy + c_n^2 \int_0^1 P_n P_m dy = 0$$

$$P_m \frac{dP_n}{dy} \Big|_0^1 - \int_0^1 \frac{dP_m}{dy} \frac{dP_n}{dy} dy + c_n^2 \int_0^1 P_n P_m dy = 0$$

using same procedure for P_m

$$\begin{aligned} & - \int_0^1 \frac{dP_m}{dy} \frac{dP_n}{dy} dy + c_n^2 \int_0^1 P_n P_m dy = 0 \\ - & - \int_0^1 \frac{dP_n}{dy} \frac{dP_m}{dy} dy + c_m^2 \int_0^1 P_m P_n dy = 0 \\ \hline & (c_n^2 - c_m^2) \int_0^1 P_m P_n dy = 0 \end{aligned}$$

Dimensional form of the equation

$$\frac{v_x(y, t)}{v_0} = \sum_{n=1}^{\infty} \left(\frac{-2}{n\pi} \right) \exp\left(-n^2 \pi^2 \frac{t}{H^2/\nu} \right) \sin\left(n\pi \frac{y}{H} \right) + \left(1 - \frac{y}{H} \right)$$

NOTE ** = velocity profile (spatial variance, for a given t)

#) In case we have condition in terms of T_0 instead of v_0 .

||||| $\rightarrow T_0$

$$\rho \frac{\partial v_x}{\partial t} = \mu \frac{\partial^2 v_x}{\partial y^2}$$

||||| $\rightarrow T_0$

we will try and convert the eqn in terms of T , using the CONSTITUTIVE RELATION

$$\uparrow$$

$$T = -\mu \frac{\partial v_x}{\partial y}$$

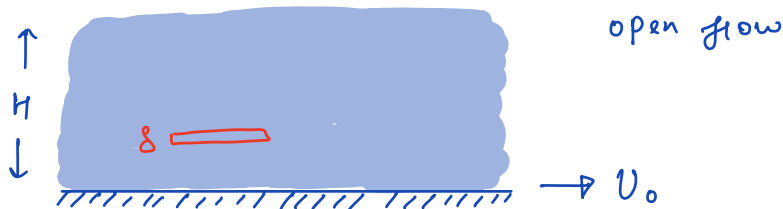
multiplying with $-\mu$

$$-\mu \frac{\partial v_x}{\partial t} = \frac{\mu}{f} \frac{\partial^2}{\partial y^2} (-\mu v_x)$$

$$\frac{\partial}{\partial y} \left(-\mu \frac{\partial v_x}{\partial t} \right) = \frac{\partial}{\partial y} \left(\frac{\mu}{f} \frac{\partial}{\partial y} \left(-\mu \frac{\partial v_x}{\partial y} \right) \right)$$

$$\frac{\partial \tau}{\partial t} = \frac{\mu}{f} \frac{\partial^2 \tau}{\partial y^2}$$

(#)



If $\delta \ll H$, semi-infinite domain.

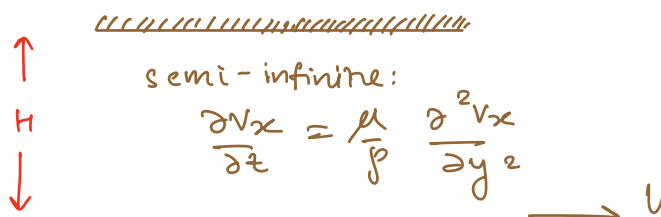
δ : region upto which fluid has moved, or momentum has been transferred.

09/02/2023

Stoke's first problem: flat plate being pulled with constant velocity



(unidirectional $v_x \neq 0, v_y = 0, v_z = 0$)



$$y=0, v_x = U, \forall t > 0$$

(Ans.)

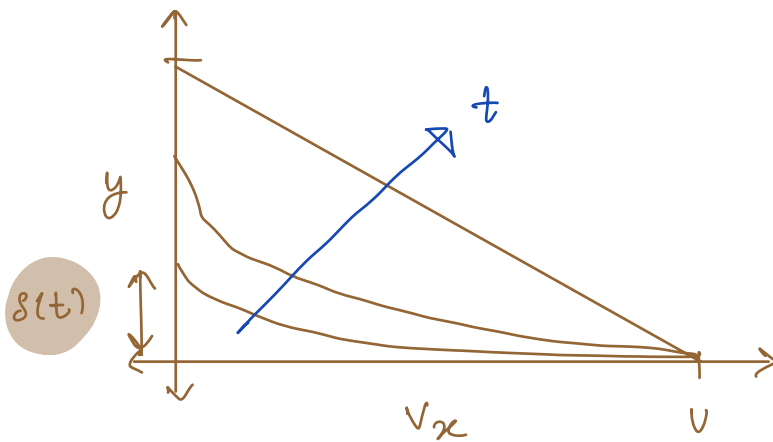
$t \sim \frac{H^2}{\nu} \rightarrow$ only if this is satisfied we can consider H

If $t \ll \frac{H^2}{\nu} \rightarrow y \rightarrow \infty, v_x = 0, \forall t$

early times

upper boundary condition
 $\rightarrow H$ will be very large as compared to the disturbance.

velocity profile:



$$t_{obs} \ll \frac{H^2}{\nu}$$

$s(t)$: characteristic length scale for early times

$$f(\eta) = \frac{v_x}{U} \quad ; \quad \eta = \frac{y}{s(t)}$$

Note: we are assuming fully developed in x-direction.

goal is to convert all variables to f & η .

$$\underline{\partial v_x} = \left(\underline{\partial v_x} \right) / \left(\underline{\partial \eta} \right)$$

$$\frac{\partial}{\partial t} \left(\frac{\partial \eta}{\partial n} \right) \left(\frac{\partial \eta}{\partial t} \right)$$

$$\frac{\partial v_x}{\partial t} = \left(v_0 \frac{\partial f}{\partial \eta} \right) \left(-\frac{\gamma}{s^2} \delta' \right)$$

$$\frac{\partial \eta}{\partial t} = (\gamma) \left(-\frac{1}{s^2} \right) \frac{d\delta}{dt}$$

$$\frac{\partial v_x}{\partial \eta} = v_0 \frac{\partial f}{\partial \eta}$$

$$\frac{\partial v_x}{\partial y} = \frac{\partial v_x}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} = v_0 \frac{\partial f}{\partial \eta} \left(\frac{1}{s} \right)$$

$$\frac{\partial^2 v_x}{\partial y^2} = \frac{\partial}{\partial \eta} \left(\frac{\partial v_x}{\partial y} \right) \cdot \frac{\partial \eta}{\partial y}$$

$$= \frac{\partial}{\partial \eta} \left(\frac{v_0}{s} \frac{\partial f}{\partial \eta} \right) \cdot \frac{\partial \eta}{\partial y}$$

$$= \frac{v_0}{s} \frac{\partial^2 f}{\partial \eta^2} \quad \downarrow \quad \frac{1}{s}$$

$$= \frac{v_0}{s^2} \frac{\partial^2 f}{\partial \eta^2}$$

ISKE BAAD PROOF DEKH LENA !!!!